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AI Quick Build : AI-Powered game-show experience with Gemini and Imagen

Zack Akil Follow 6 min read · Oct 28, 2024

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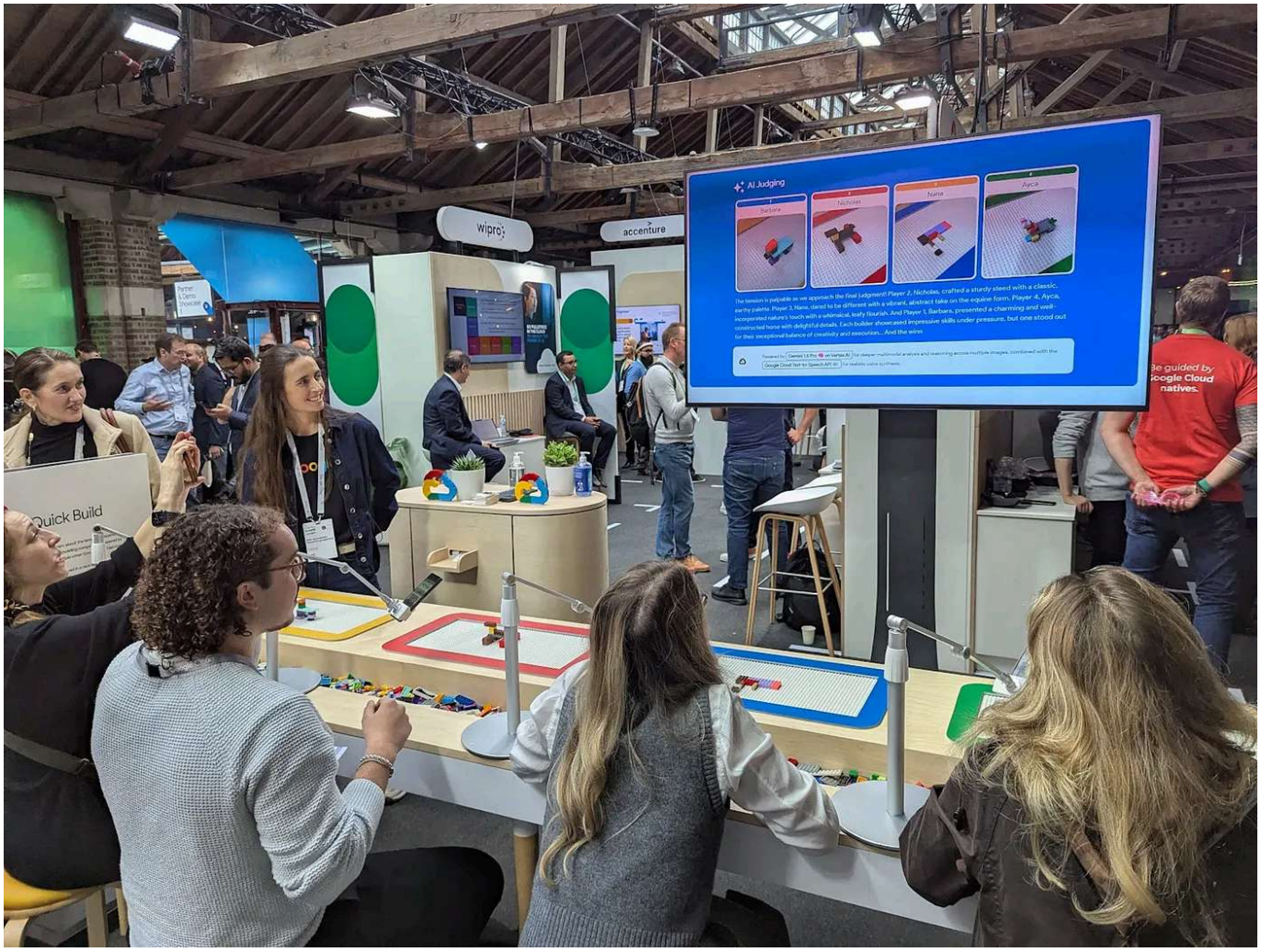
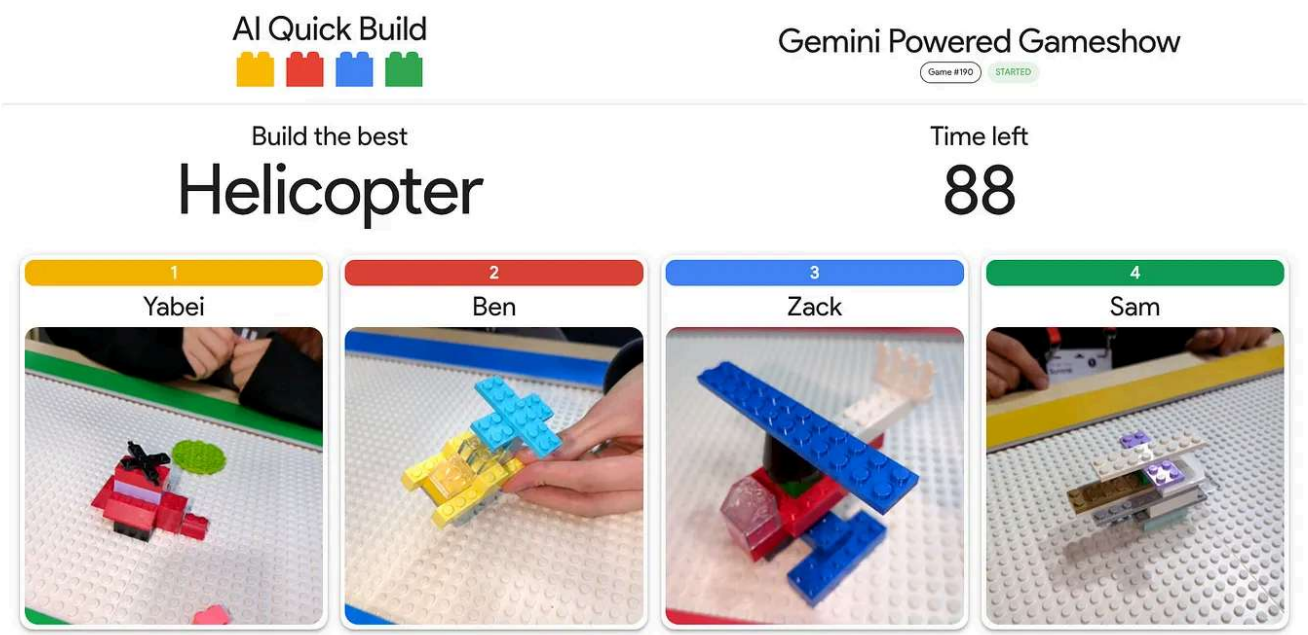


AI Quick Build @ Google Cloud London Summit 2024

Imagine a game-show where contestants race against the clock to build the most creative interpretations of different objects, all while a witty AI commentator provides live play-by-plays and judges the final creations. This isn't a scene from a futuristic movie; it's the AI Quick Build, a new demo experience powered by Google's Gemini and Imagen. This blog post takes you behind the scenes of this AI powered game-show, exploring how to combine different AI models on Google Cloud to create a truly unique and engaging experience.

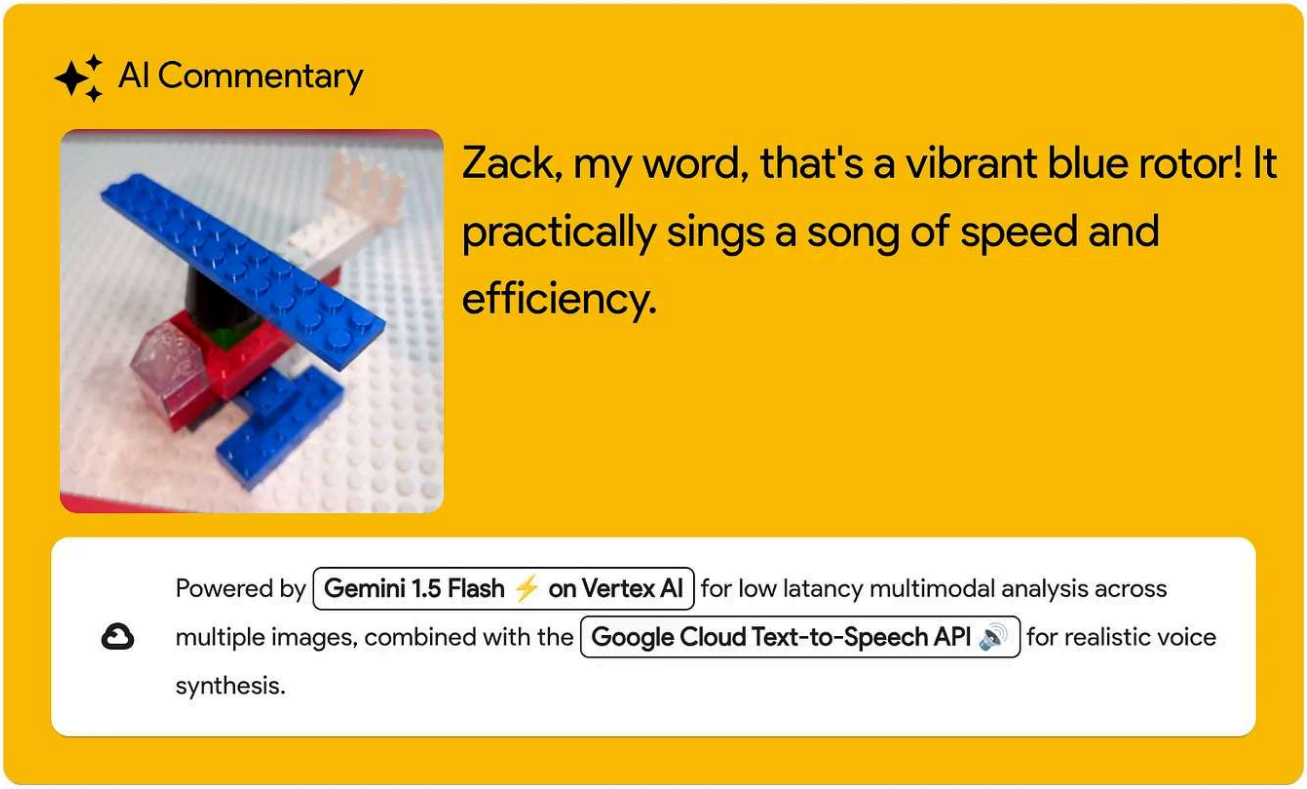
AI Quick Build: A Multi-modal Spectacle

The AI Quick Build Challenge is designed to showcase the potential of real-time multi-modal AI. Participants are given a simple prompt, like “helicopter” or “lighthouse”, and 100 seconds to bring their vision to life using toy bricks.



Throughout the building process, a Gemini-powered AI commentator analyzes the builds in real-time, generating insightful and often humorous commentary on the players’ creative choices. Once the clock stops, Gemini Pro steps in as the judge, performing a deep analysis of each creation based on specific criteria like creativity, use of color, and detail complexity. It then delivers a dramatic winner announcement script, revealing the champion and providing detailed reasoning for its decision. Finally, the builds are brought to life with Imagen, which generates a dynamic background for the creations.

Gemini Flash: The Real-Time Commentator



The heart of the experience is Gemini 1.5 Flash, a fast multi-modal modal, enabling near-real-time analysis and generation. We’ve designed a specific persona for our commentator: whimsical, but with a dash of playful snark. This persona is infused into the prompts we use to guide Gemini 1.5 Flash’s commentary. For example:

```
commentator_prompt = '''
You are a commentator at a high-stakes toy brick building contest where particip
have only 100 seconds to construct a {design}.

They will be judged on the following:

Creative Use of Color
Structural Integrity
Adherence to Brief
Detail and Complexity

Make a short comment about one of the players' builds so far.

Your persona is "Whimsical with a dash of playful snark ".
```

```
Here are the latest photos of all of the builds:
'''
```

We then append images of the builds directly into the prompt, leveraging Gemini 1.5 Flash's multi-modality to generate commentary that is both insightful and visually relevant.

Gemini 1.5 Pro: The Game-show Judge

When the building frenzy ends, Gemini 1.5 Pro takes over the judging duties.

We use a prompt that outlines the judging criteria and instructs the model to provide both detailed reasoning for its decision and a winner announcement script:

```
game_judge_prompt = '''
You are judging a high-stakes toy brick building contest where
participants have only 100 seconds to construct a {design}.

Evaluate each entry on a scale of 1-10 based on the following criteria:

  Creative Use of Color: Assess how intentionally and effectively
  different colored bricks have been used to enhance the overall
  design and convey specific themes or emotions.

  Structural Integrity: Consider the stability and durability of the build.
  Does the structure hold together under scrutiny, and does it demonstrate
  a sound understanding of brick mechanics?

  Adherence to Brief: Evaluate how closely the design aligns with the given prom
  Does it capture the essence of the concept and avoid unnecessary deviations?

  Detail and Complexity: Assess the level of detail and complexity incorporated
  Do the smaller elements contribute to the overall aesthetic and functionality?
```

```
    And choose an overall winner!

    Output an "annoucment script" that builds up to announcing the winner.
    And include a "winner reasoning" that goes into specific details about
    why that winner was choosen. Keep it consice.
    Here are the finished builds:
  '''
```

This is followed by images of each creation, allowing Gemini 1.5 Pro to provide a comprehensive and engaging judgement.

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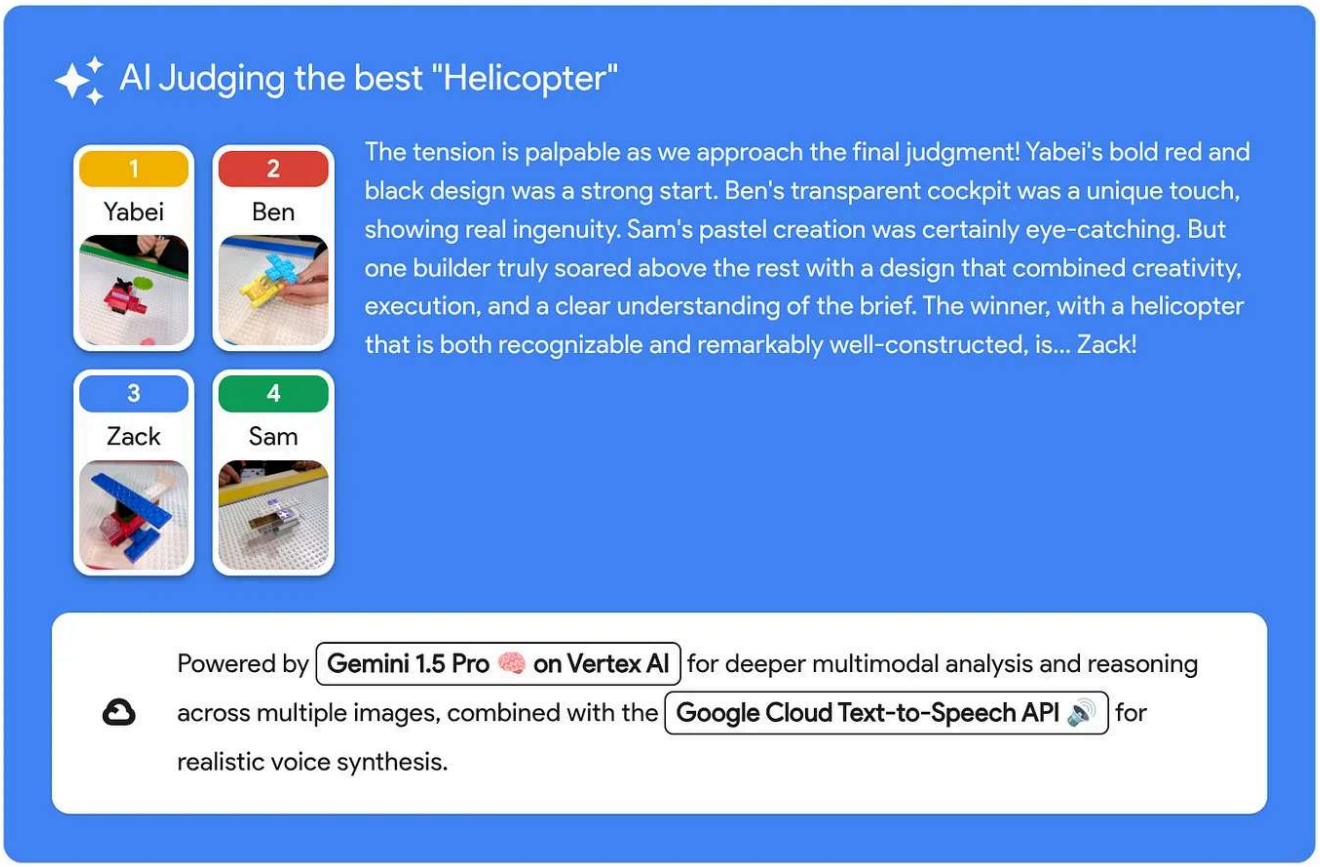
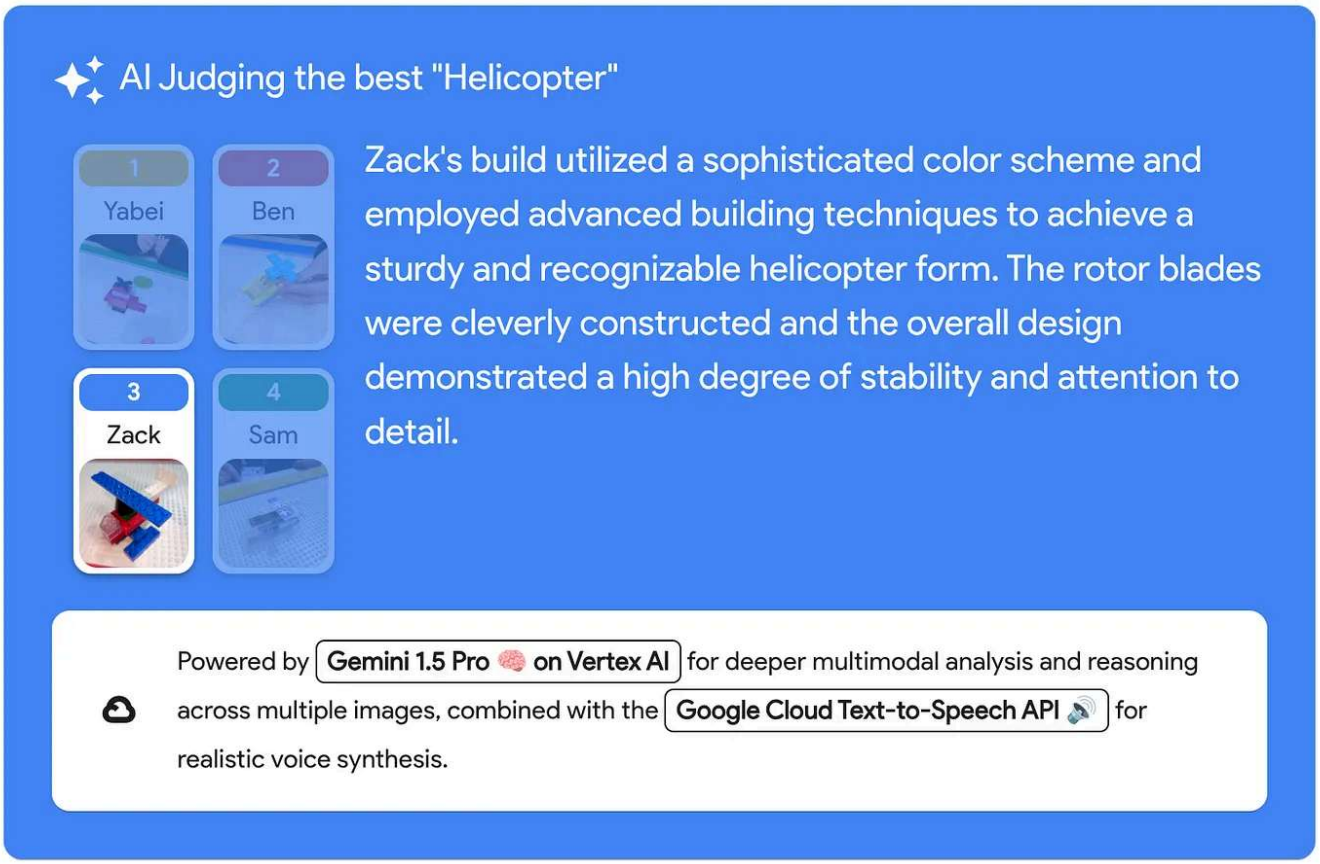
Here's what the output looks like to the contestants:



Gemini building up to the winner



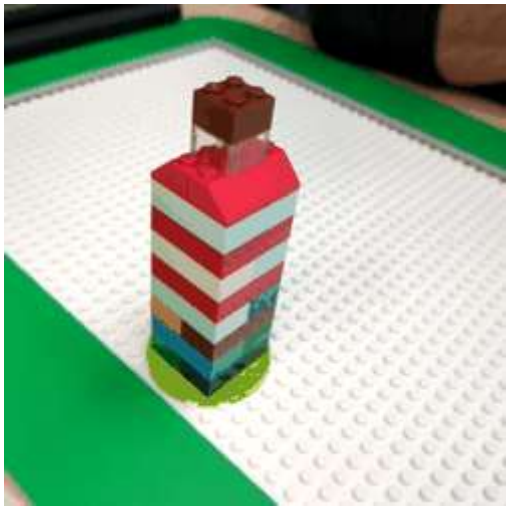
Gemini gives explanation for the winner

Giving Voice to AI with Cloud Text-to-Speech

To make the experience even more engaging, we use Google Cloud's Text-to-Speech API to give our AI commentator a voice. We leverage the latest casual voice, which injects a natural and engaging tone into the commentary. What's even more impressive is the ability to leverage punctuation to convey different emotions — all caps and exclamation marks translate into excitement, while lowercase and ellipses can convey a more thoughtful tone with this new voice. Listen to an example of this voice and others on the Cloud Text-to-Speech voice list page.

Imagen: Bringing the creations to Life

The final touch of magic comes from Imagen, Google's image generation model. We use Imagen to create a dynamic background for the winning creation, bringing it into a more imaginative context. This could be anything from a stormy seascape for a lighthouse to a lush jungle for a tiger. We then create an animated GIF that transitions between the original build and the Imagen-generated scene, providing a memorable takeaway for the contestants. If you want to see the code for how we do this check out this Python notebook on the Google Cloud Generative AI repo.



GIF showing Imagen generated background

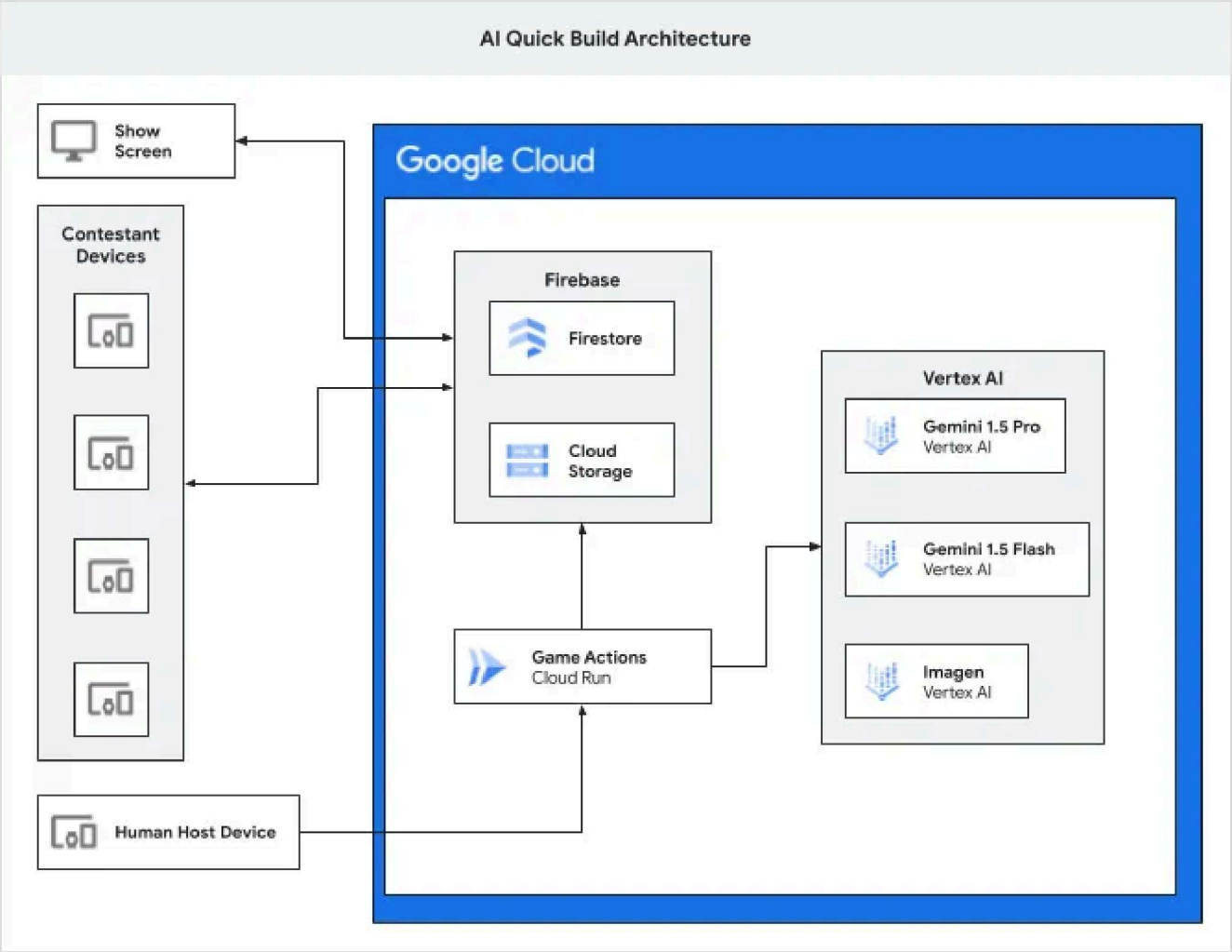
Beyond the Game-show: Applications and Possibilities

The techniques employed in the AI Quick Build can be applied to a wide range of fields:

- **Education:** Imagine a classroom where students receive personalized, real-time feedback on their work, whether they're practicing a musical instrument, perfecting a dance move, or crafting a pottery masterpiece.
- **Interactive Entertainment:** This technology opens the door to a new breed of games and immersive experiences, where AI commentators and judges provide engaging feedback and dynamic story lines based on player actions.
- **Nuanced Visual Analysis:** Consider industries like construction or manufacturing, where visual inspections are crucial. Multi-modal models like Gemini can be quickly adapted to analyze the correctness of tasks based on images and prompts, providing a powerful tool for quality control and training.

Built on Google Cloud

The entire AI Quick Build experience is built on the scalable infrastructure of Google Cloud, leveraging Cloud Run, Firebase, and Vertex AI. For developers interested in diving deeper into the technical aspects of the back-end, we've published a Jupyter Notebook that outlines the Python code for interacting with Gemini and Imagen.



Google Cloud Architecture of AI Quick Build

Key takeaways

The AI Quick Build is a demonstration of how you can build real-time multi-modal generative AI applications on Google Cloud. Specific tool highlights from the project are:

- **Gemini Flash** provides dynamic, real-time commentary, adapting to the visual input of the builds.
- **Gemini Pro** acts as a comprehensive game judge, offering detailed reasoning for its decisions.
- **Imagen** elevates the experience by generating imaginative backgrounds, bringing the creations to life.

Look out for the AI Quick Build at an upcoming Google event!

Check out the Python notebook to see the back-end code in action:

generative-ai/gemini/use-cases/education/ai_quick_build_experience_backend.ipynb at...

Sample code and notebooks for Generative AI on Google Cloud, with Gemini on Vertex AI ...

github.com

More resources:

- Gemini documentation
- Imagen documentation
- Vertex AI Generative AI

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Written by Zack Akil

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Hello Zack - I played this game today @GTC. Could you help me with UI part any link/doc? I already reviewed the backend python code.

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